Cursul 9 - Fizică

1. Funcție de distribuție MAXWELL: după mărimea vitezei:
   - 1.1. viteza medie
   - 1.2. viteza pătratică medie
   - 1.3. viteza cea mai probabilă
2. Aplicații. Energia medie a unui oscilator cuantic

$$E_n = \hbar\omega\left(n+\frac{1}{2}\right);\quad n=0,1,2,\dots$$

① $v \div v+dv \Leftrightarrow p \div p+dp$

$$P(A) = \frac{1}{Z}\int_{-\infty}^{\infty}\dots\int_{p_i}^{p_i+dp_i}\int_{-\infty}^{\infty} e^{-\frac{p_i^2}{2mkT}}\,dx_N$$

$$dx_N = \frac{1}{h^{3N}\cdot N!}\,d\vec{r}_1\cdot\dots\cdot d\vec{r}_n\cdot d\vec{p}_1\cdot\dots\cdot d\vec{p}_n$$

$$Z = \int_{-\infty}^{\infty}\dots\int_{-\infty}^{\infty} e^{-\frac{p_i^2}{2mkT}}\,dx_N$$

$$P(A) = \frac{\int_{p_i}^{p_i+dp_i} e^{-\frac{p_i^2}{2mkT}}\,d\vec{p}_i}{\int_{-\infty}^{\infty} e^{-\frac{p_i^2}{2mkT}}\,d\vec{p}_i}$$

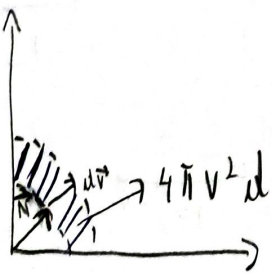
$$\int_{-\infty}^{\infty}\iint e^{-\frac{p_{ix}^2+p_{iy}^2+p_{iz}^2}{2mkT}}\,dp_{ix}\,dp_{iy}\,dp_{iz}$$

$$\left[\int_{-\infty}^{\infty} e^{-\frac{p_{ix}^2}{2mkT}}\,dp_{ix}\right]^3 \longrightarrow \int_{-\infty}^{\infty} e^{-dx^2} = \sqrt{\frac{\pi}{\alpha}}$$

$$P(A) = \frac{\int_{v_i}^{v_i+dv_i} e^{-\frac{mv_i^2}{2k_BT}}\,d\vec{v}}{\left(\frac{2\pi k_B}{m}\right)^{3/2}} = \left(\frac{m}{2\pi k_BT}\right)^{3/2}\int_{v_i}^{v_i+dv_i} e^{-\frac{mv_i^2}{2k_BT}}\,d\vec{v}_i$$

$d\vec{v}_i \downarrow 4\pi v^2 dv$

$$V = \frac{4}{3}\pi v^2 \Rightarrow dV = 4\pi v^2 dv$$



1.1. $\bar{v} = \int_0^\infty v\, f(v)\, dv = 4\pi \left(\frac{m}{2K_B\bar{T}}\right)^{3/2} \int_0^\infty v^3 e^{-\frac{mv^2}{2K_B\bar{T}}} dv$

$I_{m,n} = \int_0^\infty x^m e^{-\alpha x^n} dx = \frac{\Gamma\left(\frac{n+1}{n}\right)}{\alpha^{\frac{n+1}{n}}}$

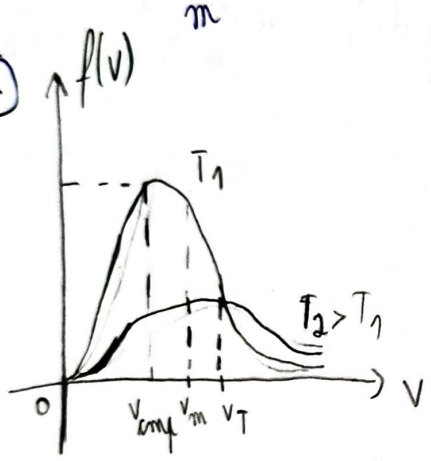
$$\boxed{\bar{v} = \sqrt{\frac{8K_B\bar{T}}{m}}}$$

1.2. $\overline{v^2} = \int_0^\infty v^2 f(v)\, dv = 4\pi \left(\frac{m}{2\pi K_B\bar{T}}\right)^{3/2} \cdot \int_0^\infty v^4 e^{-\frac{mv^2}{2K_B\bar{T}}} dv$

$I_1 = \int_0^\infty x^2 e^{-\alpha x^2} dx = -\frac{d I_0}{d\alpha} = -\frac{d}{dx}\left(\sqrt{\frac{\pi}{\alpha}}\right)$

$\overline{v^2} = \frac{3K_B\bar{T}}{m}$ $\qquad v_T = \sqrt{\overline{v^2}} = \sqrt{\frac{3K_B\bar{T}}{m}}$ viteza termică

1.3.



v_{cmp} = viteza cea mai probabilă

$\left[\frac{df}{dv}\right]_{v=v_{cmp}}$ pt. aflarea maximului

$f(v) = 4\pi v^2 \left(\frac{m}{2\pi K_B\bar{T}}\right)^{3/2} e^{-\frac{mv^2}{2K_B\bar{T}}} \Rightarrow 2v e^{-\frac{mv^2}{2K_B\bar{T}}} + v^2\left(-\frac{2mv}{2K_B\bar{T}}\right) e^{-\frac{mv^2}{2K_B\bar{T}}} = 0$

$$\Rightarrow \boxed{v_{cmp} = \sqrt{\frac{2K_B\bar{T}}{m}}}$$

$f(v)\, dv = \frac{dN}{N} \Rightarrow dN = N f(v)\, dv$

② a) funcția de poziție

$Z = \sum_{n=0}^{\infty} e^{-\frac{E_n}{K_B\bar{T}}} = e^{-\frac{\hbar\omega}{2K_B\bar{T}}} \cdot \underbrace{\sum_{n=0}^{\infty} e^{-\frac{n\hbar\omega}{K_B\bar{T}}}}_{\sum_{n=0}^{\infty} e^{-nx} = \frac{1}{1-e^{-x}}} = e^{-\frac{\hbar\omega}{2K_B\bar{T}}} \cdot \frac{1}{1 - e^{-\frac{\hbar\omega}{K_B\bar{T}}}}$

temă: $Z = \frac{1}{2\,\mathrm{sh}\left(\frac{\tilde{\hbar}\omega}{2K_BT}\right)}$

b) energia medie:

$$\bar{E} = U = k_B \cdot T \cdot \frac{\partial(\ln Z)}{\partial T} = k_B \cdot T \cdot \frac{1}{Z} \cdot \frac{\partial Z}{\partial T} = \ldots = -\frac{\tilde{\hbar}\omega}{2}\,\mathrm{cth}\left(\frac{\tilde{\hbar}\omega}{2K_BT}\right)$$